Natural isomorphism

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Recall the definition of adjoint functors,

Functors $F : \boldsymbol{C} \to \mathbf{D}, G : \mathbf{D} \to \boldsymbol{C}$ are called adjoints of each other iff

$$\forall c \in \mathrm{Obj}(\boldsymbol{C}), d \in \mathrm{Obj}(\mathbf{D}), \mathrm{Hom}_{\boldsymbol{C}}(c, G(d)) \cong \mathrm{Hom}_{\mathbf{D}}(F(c), d),$$

and the isomorphism is **natural** in c and d.

Now, observe this definition of representable multicategory,

> 8.1. Definition. A multicategory $\mathbb{M}$ is said to be *representable* if
>
> (1) for every tuple of objects $\vec{x} = \langle x_1, ..., x_n \rangle$ there exists a *universal arrow*, $\pi_{\vec{x}} : \langle x_1, ..., x_n \rangle \to \otimes \vec{x}$, in the sense that precomposition with this arrow induces an isomorphism
>
> $$\mathbb{M}(\langle x_1, ..., x_n \rangle, y) \cong \mathbb{M}(\langle \otimes \vec{x} \rangle, y)$$
>
> natural in y, and
>
> (2) universal arrows are closed under composition.

Both definitions talk of an isomorphism being natural in some parameters.

Consider two functors $F, G : \boldsymbol{C} \to \mathbf{D}$, and natural transformations $\tau_1 : F \to G, \tau_2 : G \to F$ such that $\tau_1 \circ \tau_2 = \mathrm{id}_G, \tau_2 \circ \tau_1 = \mathrm{id}_F$.

Then we say that for any $y \in \mathrm{Obj}(\boldsymbol{C})$, $F(y)$ and $G(y)$ are isomorphic (via the morphisms $\tau_1(y)$ and $\tau_2(y)$), and the isomorphism is natural in the parameter y.